

SageCam v1.0 with SageMic add-on

Conservation Technology Lab

San Diego Zoo Wildlife Alliance



[Bill of Materials](#)

[Wiring Diagram](#)

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1. Materials

There are tools needed to manufacture the box, as well as other tools needed to deploy the unit, and software packages needed to configure some of the components.

1.1 Tools - Building

Drill press

Scissors

Small Philips screwdriver

Small flathead screwdriver

Wire strippers

Ethernet crimp

Pliers

Wrench

Silicon glue

Soldering iron*

Optional* To help with adhering the o-terminals to the battery wires

1.2 Tools - Deployment

Battery powered hammer drill

Mallet

2x Frame Packs

Bungee/Straps

Nippers

1.3 Tools – Software

Raspberry Pi Imager version $\geq 2.0.3$ - computer

AudioMoth Firmware Flash App - computer

AudioMoth Microphone Configuration App - computer

UISP App - phone

1.4 Hardware

See the [BOM](#) for a complete list of hardware requirements and suggested vendors.

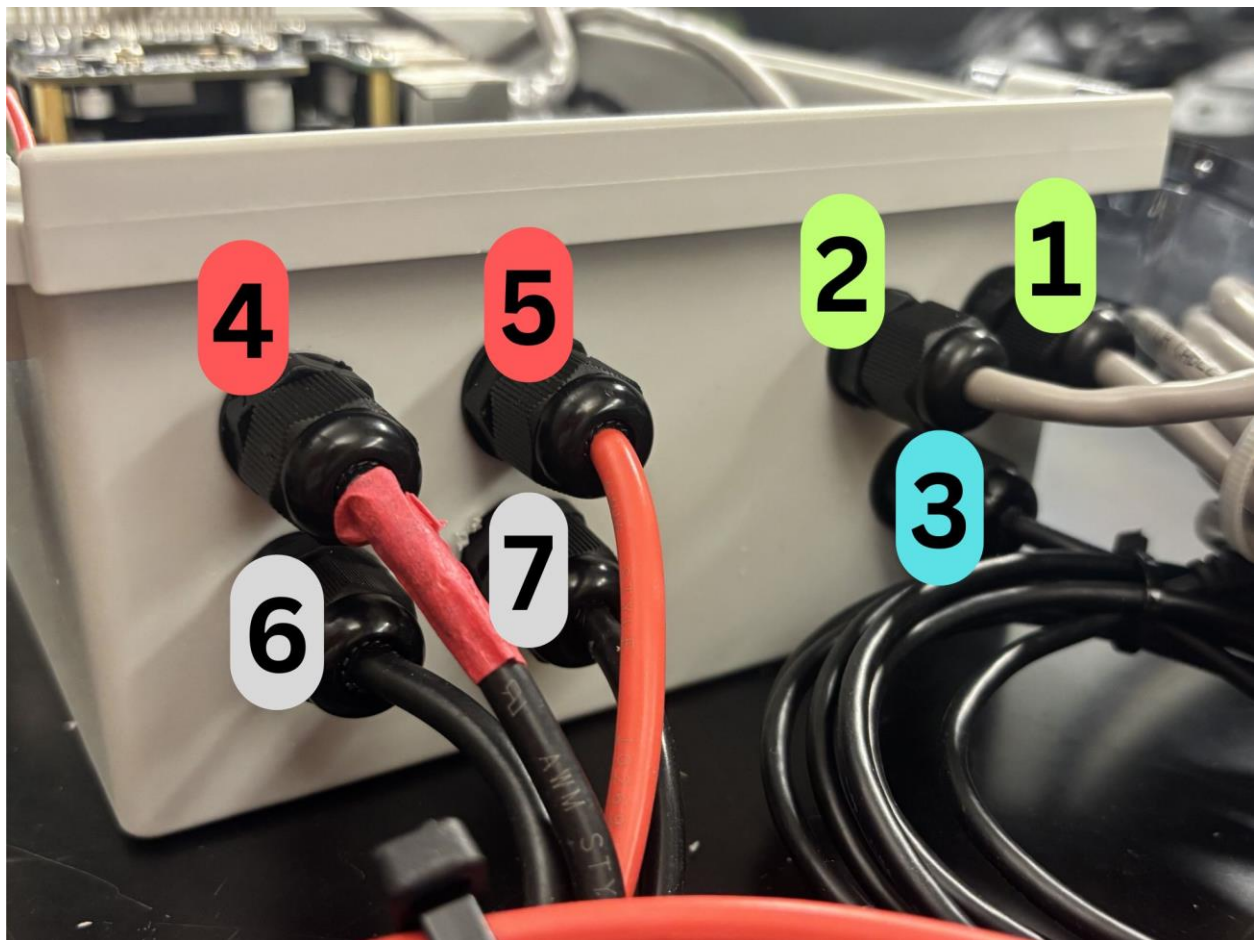
2. Box

2.1 Drilling holes

7x Cable Glands

For a Nanobeam-connected sagecam with the sagemic addition, 7 holes need to be drilled into the electrical box.

Place cable glands in each hole to ensure fit, do not tighten too tightly because they will be loosened as we will thread cables through each one.



2.2 Making ethernet cables

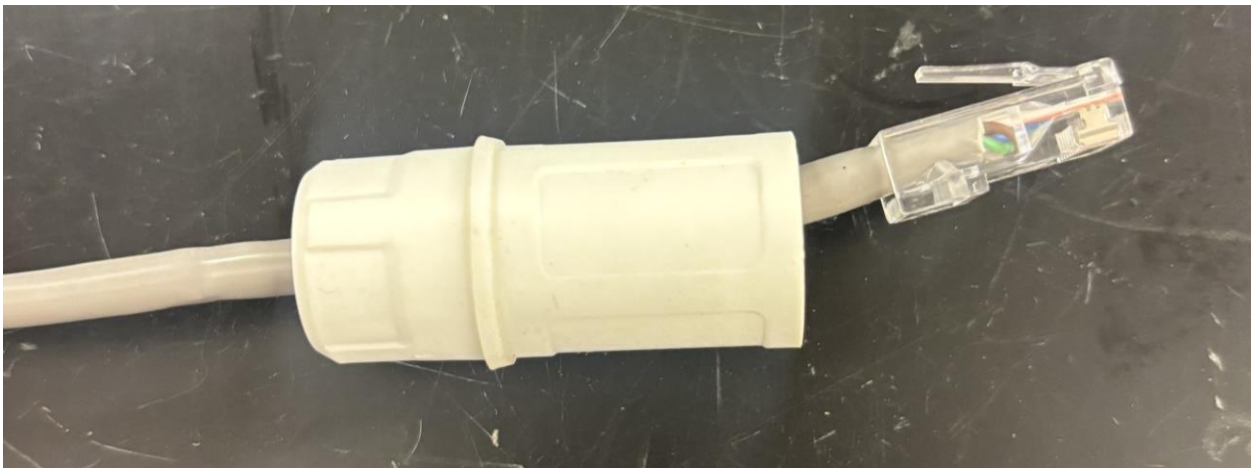
2x 12ft Ethernet Cables

2x 6in Ethernet Cables

4x Ethernet Connectors

1x PTZ Cable Gland

- 1) Cut 2x 12 ft long Cat-6 ethernet cable.
- 2) Thread one end through cable gland 1, and the other through cable gland 2.
- 3) Add ethernet connectors to both ends INSIDE the boxes.
- 4) Add an ethernet connector to ONE of the outside ends.
- 5) You should have 1 ethernet cable end outside the box without a connector, ensure to add the cable gland that came with the PTZ camera onto the ethernet cable BEFORE adding the ethernet connector.



- 6)
- 7) Create 2 additional 6 inch ethernet cables, or use some from off the shelf. These will be inside the box, one going from the Raspberry Pi hat -> the 3 port switch, and 1 going from the ubiquiti poe injector -> the 3 port switch.

2.3 Making Battery/Solar cables

2x 6ft Black cable

2x 6ft Red cable

6in red 20 guage cable

6in black 20 guage cable

2x 10mm inner diameter o-ring terminals

Male/Female Solar Connectors

- 1) Cut 2x 6ft of black cable, and 2x 6ft of red cable.
- 2) On 1 black and 1 red end, strip and crimp on an o-ring connector. If using the battery we suggest in the BOM, the o-terminals will need to have a 10mm interior

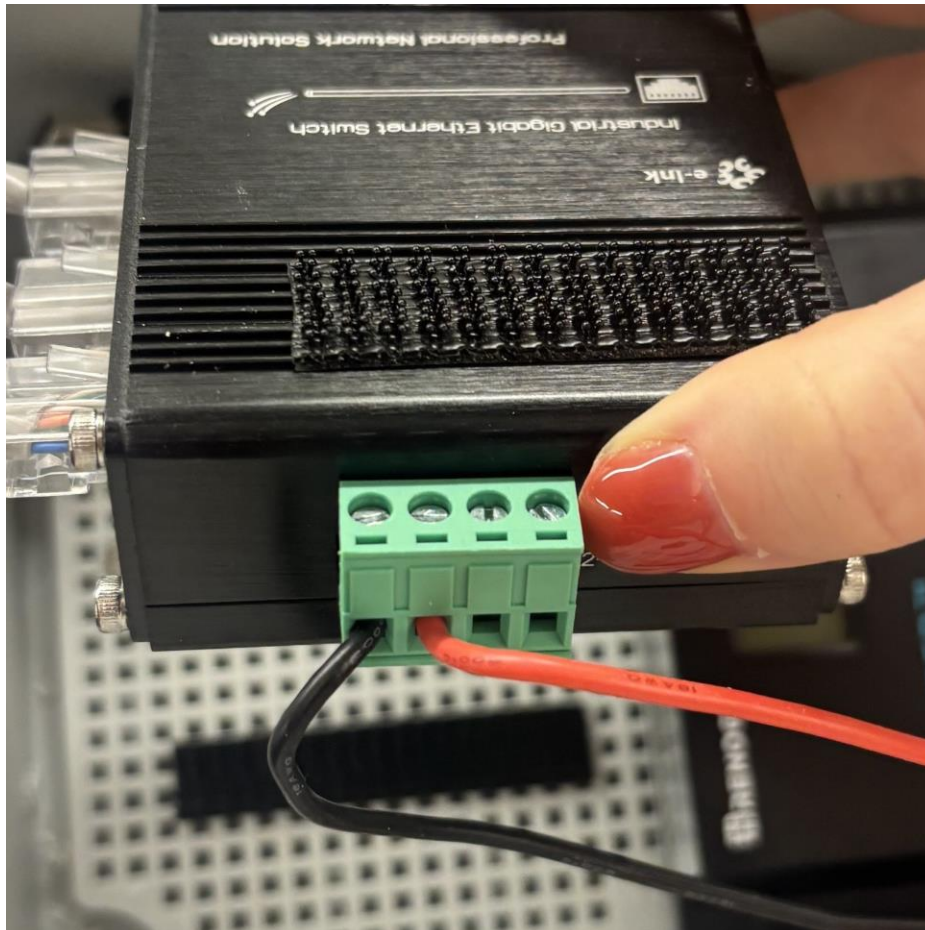


diameter.

- 3) Optionally you can solder the tips for added ruggedization.
- 4) Strip about ½ inch off the other 2 ends and insert into cable glands 5 and 7 on the image above, red on 5 and black in 7. We will insert these ends into the solar manager later.
- 5) Do the same with the other black and red wire, black through 6 and red through 4.
- 6) Attach solar panel connectors onto the other ends, the male terminal on the red wire and the male connector on the black wire.



- 7) Strip both ends of the red and black 6 inch smaller cables. Insert one end of each into the 3 port active poe switch power block. Leave the other 2 ends loose for now, they will go into the solar power manager later.



2.4 Making USB -> AudioMoth connections

MicroUSB -> USB Cable

Electrical Box

Acoustic Adhesive Membrane

Foam Tape

The MicroUSB cable needed to connect the AudioMoth to the raspberry pi is too large on either end to fit through the cable gland. This means we need to cut the usb cable, fit it

through the cable gland, and reattach it to itself so that the usb is inside the box and the microusb is outside the box.

- 1) Cut the microusb cable, cutting a few inches from the USB end.
- 2) Insert through the cable gland so the microusb side is outside the box, and the cut end is inside the box.
- 3) Solder each wire back to its original wire on the usb end inside the box, using heat shrink tubing around each connection point so they don't short each other.
- 4) Use a larger heat shrink tubing around the outside of all the connection points.
- 5) You should now have a microusb -> usb cable where the usb cable is inside the box, along with the soldered connections, and the microusb and intact cable is extending outside the box.
- 6) For the AudioMoth, we manufacture a custom casing using a small electrical box.
- 7) Cut a small hole where the microphone will sit when inside, and use cushion tape to make the AudioMoth fit snugly within the case so that the microphone stays in place.



- 8) Using an acoustic filter sticker, cover the hole from the inside.
- 9) Drill a hole large enough to fit the microusb cable and plug the AudioMoth in.
- 10) Screw the case shut and use silicon glue around the microusb cable hole to make it waterproof, leave to cure at least overnight.

2.5 Raspberry Pi hat/casing

Raspberry Pi

Raspberry Pi Case

POE Pi Hat

- 1) We will only be using the bottom plate from the raspberry pi casing kit, so attach that to the bottom of the pi using the screws that come with it.
- 2) Attach the poe hat on top of the pi.
- 3) Done!

2.6 Placing components

7x Strips of velcro (14 total)

3 Port POE Injector

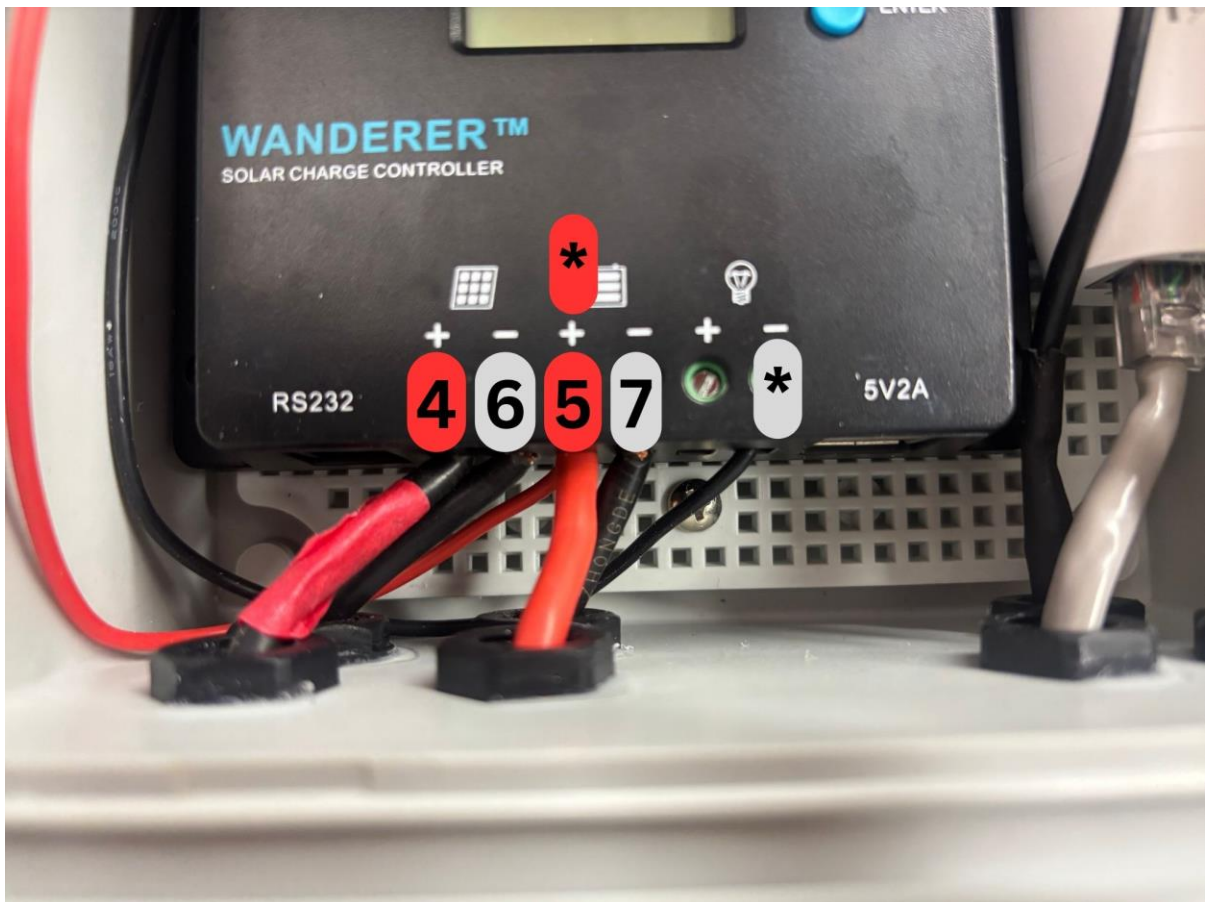
Solar Charge Controller

Ubiquiti POE injector

Pi with Casing/Hat

2x 6in ethernet cables (from step 2.2)

- 1) Using the screwdriver, loosen the terminals on the solar power manager and insert the solar, battery, and power cables in the proper terminals. Refer to the first image to correlate the numbers.



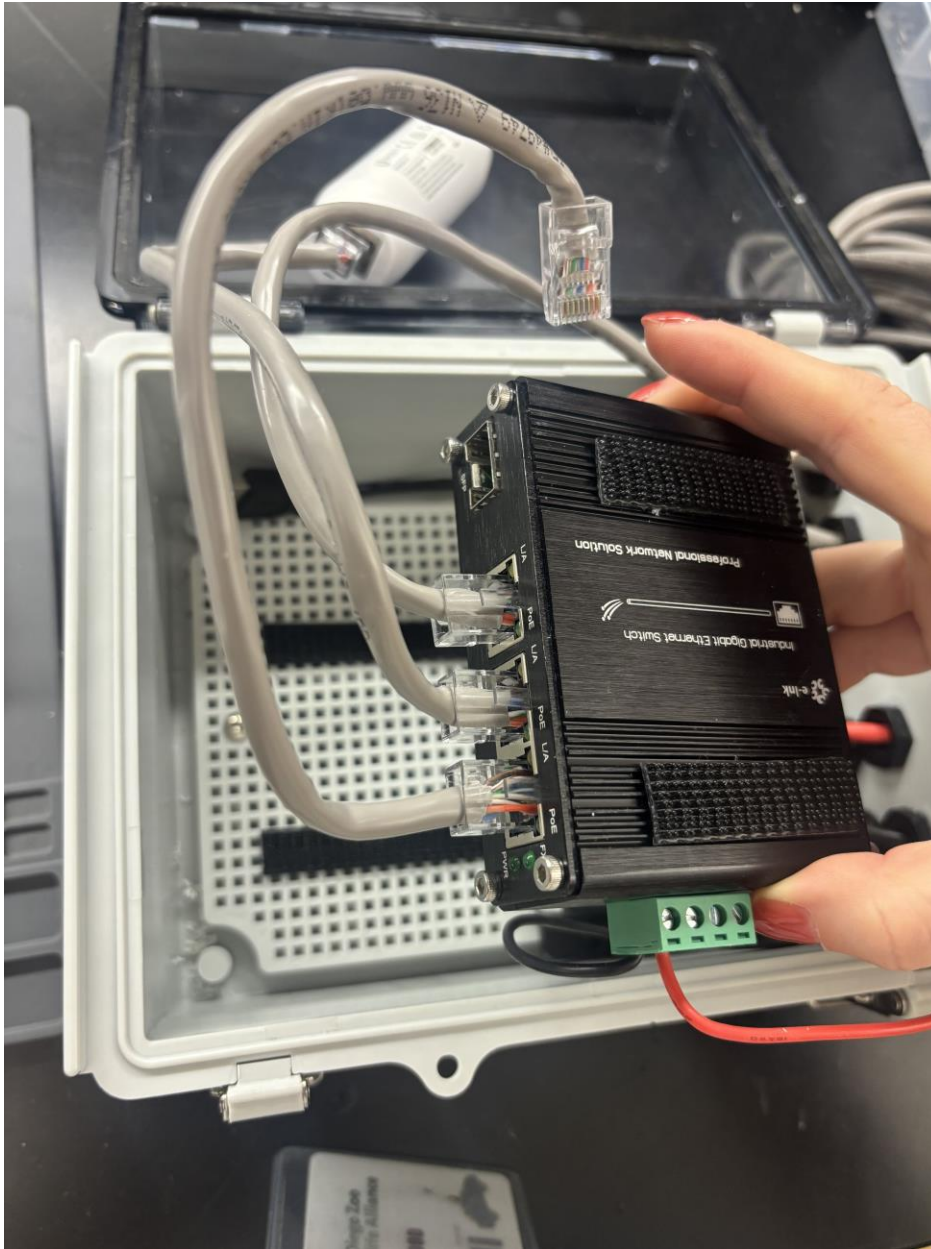
- 2) The asterisks in the image denote the positive and ground small 20 guage wires coming from the 3 port active poe switch from earlier, the positive terminal will go

inside the battery power alongside the battery connector. Ground will be in the lightbulb negative terminal.

- 3) Use 2 strips of velcro to attach the solar manager to the electrical box.



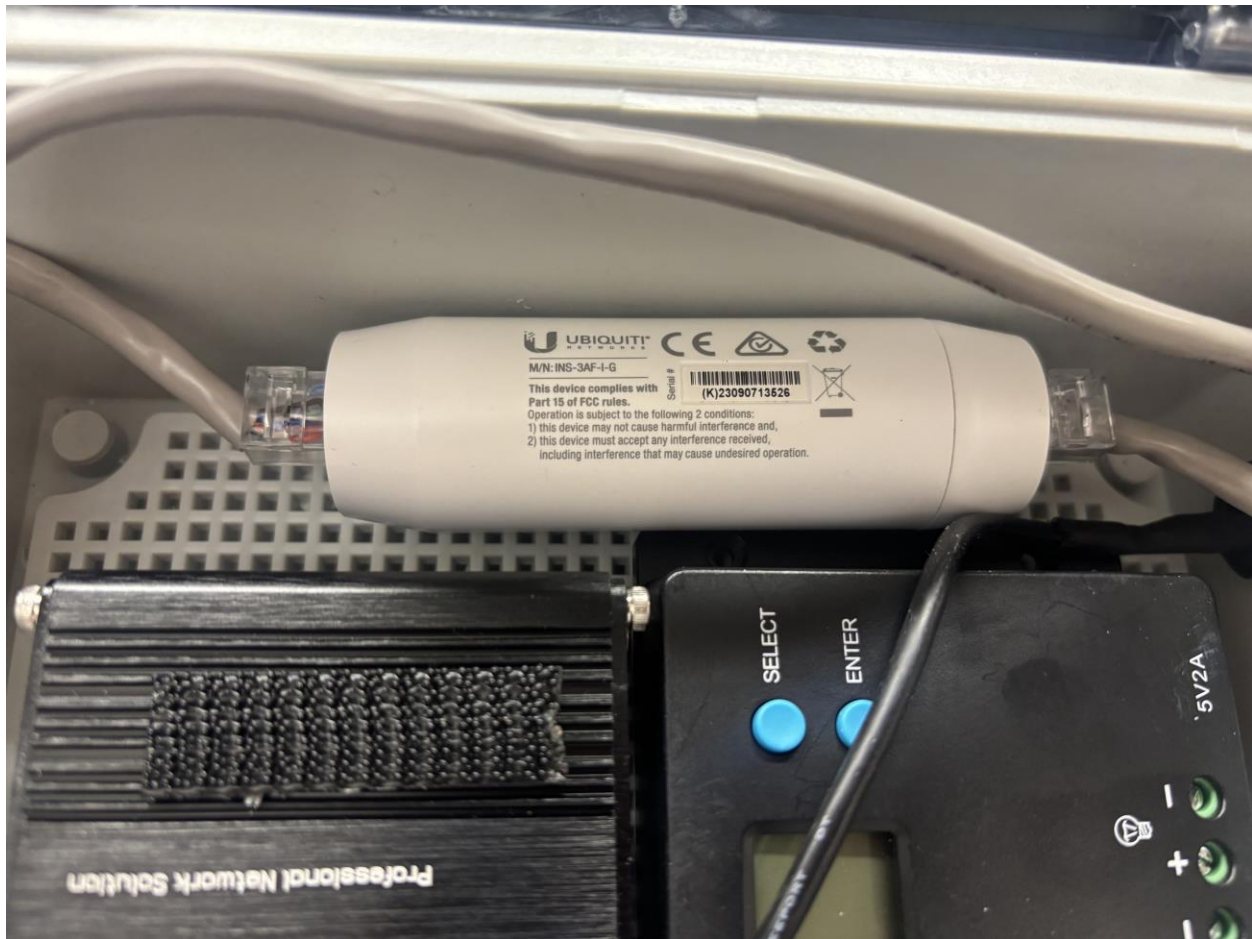
- 4) Insert the 3 ethernet cables into the 3 port switch before attaching with velcro to the bottom of the box.



- 5) Add 2 more pieces of velcro on top of the 3 port switch, and attach the raspberry pi on top.



- 6) Finally, attach the ethernet cable that does NOT have the camera cable gland into the POE OUT side of the ubiquiti passive poe injector. Insert the internal ethernet cable connected to the switch to the other side. Place inside the open space on top of the switch/solar manager. You can use an additional piece of velcro to secure it.



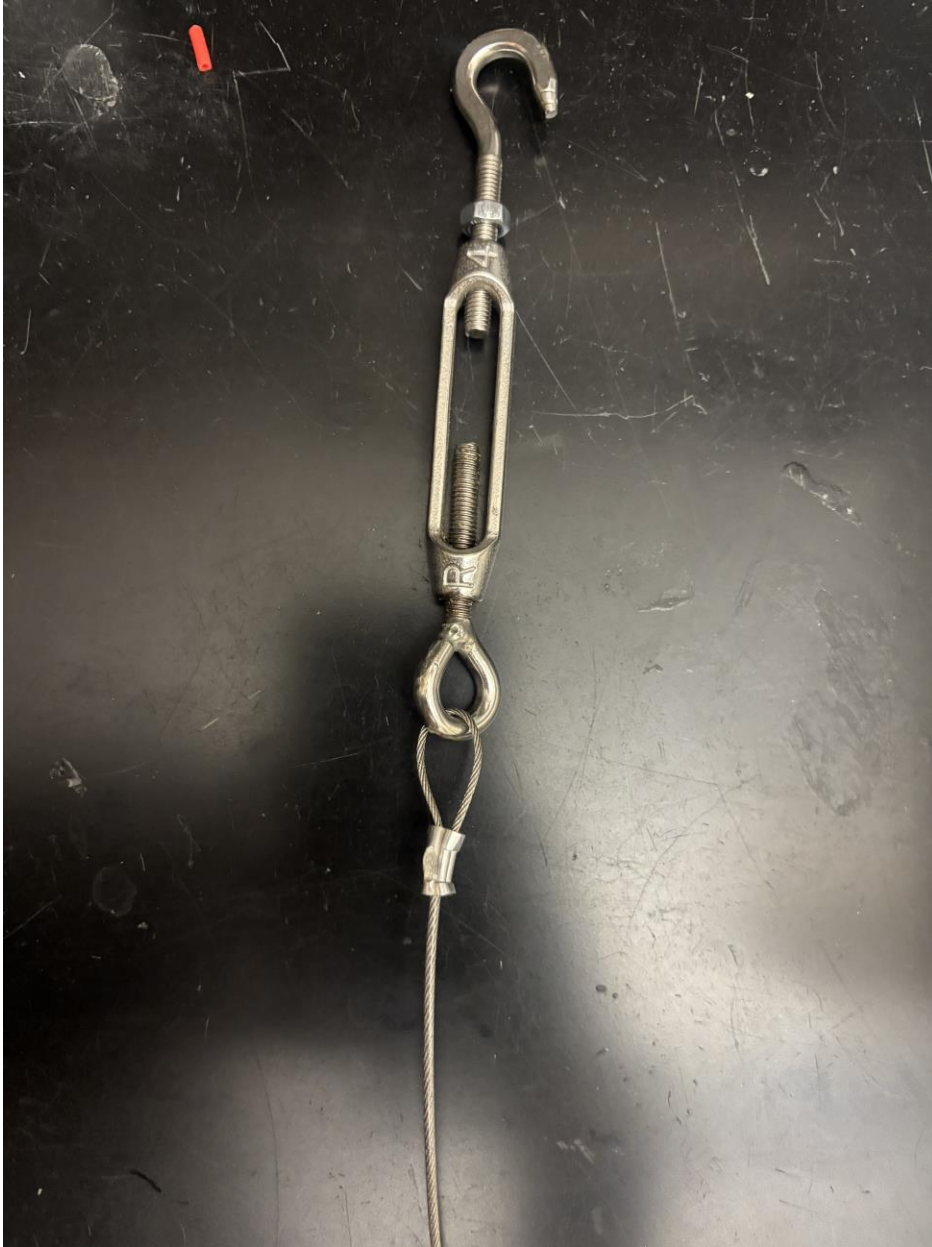
2.7 Guy-lines

3x 15ft metal wire

3x turnbuckles

3x stakes

- 1) Cut 3 strips metal cable at 15ft each.
- 2) Crimp loops on one end of each, creating a loop of about 3 inches in diameter.
- 3) Crimp the other end to the closed eye-bolt of the turnbuckle.



- 4) The open hook end will be what grabs onto the metal loop on the ground stake.

3. Software

The Nanobeam, Raspberry Pi, and Audiomoth must be configured prior to deployment.

3.1 Nanobeam

- 1) When powered on, the Nanobeam will activate a wifi to join to allow for configuration for the first time, join the wifi and navigate to 192.169.230.1

- 2) After initial setup, ensure the firmware version matches the one on the beam that this nanobeam will connect to. If it doesn't, either update both to the most recent firmware or download the specific release to update the current nanobeam being configured.
- 3) Set a static IP address that matches the same subnet of the beam it will be connected to and ensure they have the same Network configuration username and password, and that they are set to the same frequency range.

Network Role

NETWORK MODE Bridge ▾

Configuration Mode

CONFIGURATION MODE Simple ▾

Management Network Settings

MANAGEMENT IP ADDRESS ☐ DHCP ☒ STATIC

IP ADDRESS 192.168.1.1

NETMASK 255.255.255.0

GATEWAY IP 192.168.1.254

PRIMARY DNS IP 8.8.8.8

SECONDARY DNS IP 1.1.1.1

MTU 1500

STP OFF

MANAGEMENT VLAN OFF

AUTO IP ALIASING ON

DHCP OPTION 82 OFF

IPV6 OFF

Above is an example network setup, please fill in your own IP address/gateway Netmask/DNS for your own network.

Basic Wireless Settings

ACCESS POINT	☐ OFF
PTP MODE	☐ OFF
CHANNEL WIDTH	20/40 MHz ▾
CONTROL FREQUENCY SCAN LIST, MHz	☐ OFF
SSID	BDRS SELECT...
LOCK TO AP MAC	

The above image will be the configuration of the nanobeam set-up on the device.

Basic Wireless Settings

ACCESS POINT	☒ ON
PTP MODE	☐ OFF
CHANNEL WIDTH	40 MHz ▾
FRAME DURATION	Flexible ▾
CONTROL FREQUENCY LIST, MHz	☐ OFF
CENTER FREQUENCY, MHz	5200 ▾
CONTROL FREQUENCY, MHz	5190 ▾
SSID	BDRS

The above image will be the configuration of the nanobeam our device nanobeam connecting to, this one will be an AP with the set frequency. You can use a different frequency than the one shown.

- 4) In a typical setup, the beam that is providing the connection will be an AP (access point) and the beam being configured on the node will not- you can insert the MAC address of the AP beam for easier deployment, as the beam will immediately lock to that AP when powered in the field.

- 5) In the images, SSID “BDRS” is shown. That is our SSID, feel free to create your own and set your own password, all Nanobeams connected to each other either as relays or connected via switch will need the same SSID and password. You may have multiple SSIDs and passwords for Nanobeam connections on the same network if they aren’t trying to connect directly to each other.

3.2 Raspberry Pi

- 1) Flash the newest raspi-OS (Trixie as of 2026) with full desktop onto the microSD card, ensure to add your wifi SSID and password and your ssh key when configuring the flash.

Customisation: SSH authentication

Configure SSH access

Enable SSH

[Learn about SSH](#)



Authentication mechanism:



Use password authentication



Use public key authentication

1 SSH key configured

[SHOW](#)

- 2) Once flashed, plug it in and turn it on.
- 3) You should be able to ssh for the first time using the hostname of the pi using .local
 - a. If it cannot resolve hostname, use a keyboard, mouse, and monitor to complete the following steps. Sometimes hostname resolution can be finnick.
- 4) Once in the pi, configure a static ethernet IP on the same subnet as the Nanobeam, there is a helpful tutorial on how to do this via the command line [here](#).
- 5) Follow the sagemic instructions here to set the pi up for streaming its audio.
- 6) Follow the sagecam tutorial here if you’d like to capture timelapses from the PTZ cam.

3.3 Audiomoth

- 1) Connect the AudioMoth (set to USB/OFF) to your laptop.
- 2) Flash the microphone firmware onto the device from the Audiomoth firmware app.
- 3) Go to the microphone configuration app and select the frequency you'd like to record at, we choose 48KHz because that is what BirdNet will expect within our SageMic software.
 - a. Note that a lower sample rate will still work and requires no changes to the sagemic firmware.
- 4) Once configured, unplug and move switch to CUSTOM.



3.4 PTZ Cam

- 1) The cameras inherent IP address is 192.168.1.64. To change this to match the network of the pi and nanobeam, turn the camera on and connect it to your computer via ethernet.
- 2) Change the ip address of your machine to match the camera, something on 192.168.1.XX. SSH into the camera, and navigate to 192.168.1.64 in your browser.
- 3) In the camera settings, change the static ip to match the ip address of your pi and nanobeam network.

Live View

Configuration

TCP/IP

DDNS

PPPoE

Port

NAT

Multicast

NIC Type

Auto

☐ DHCP

IPv4 Address

192.168.1.2

✓

Test

IPv4 Subnet Mask

255.255.255.0

IPv4 Default Gateway

192.168.1.254

✓

IPv6 Mode

Route Advertisement

View Route Advertisement

IPv6 Address

IPv6 Subnet Mask

IPv6 Default Gateway

::

Mac Address

bc:ba:c2:64:ba:46

MTU

1500

☒ Enable Multicast Discovery

DNS Server

Preferred DNS Server

8.8.8.8

Alternate DNS Server

1.1.1.1

Save

This is an example network setup, ensure to use your own network IP address- on the same subnet as your nanobeam and pi.

- 4) You will lose connection to the camera once this is set, to check if it was configured properly, temporarily change the ip address of your machine to also be on the same network as the ptz cam, pi and nanobeam.
- 5) Navigate to the newly set ptz ip on the browser, you should see the admin page pop up, if it does, you have correctly configured the ip address of the camera.

*When setting the PTZ cam password, we recommend not using special characters.

When interacting with the PTZ cam programmatically, special characters in the password can introduce strange behavior.

3.5 Solar Charge Controller

If using the recommended Lithium battery, you'll need to change the default setting on the solar charge controller to handle lithium-type battery instead of Gel.

- 1) While the device is powered on (connected to the battery), navigate to the battery display, where you can see the battery voltage.
- 2) Hold down "Select" until you see the display blinking, you should see the default preset "Gel"
- 3) Press "Enter" until you see "Li" blinking on the screen.
- 4) Leave it once "Li" is set, it will take about 10 seconds to go back to the regular screen and it will adopt the Lithium setting.

4. Deployment

Select a spot for your deployment- ideally on relatively flat ground, with a line-of-sight to the nearest Nanobeam.

4.1 Packing for the trip

The better you pack, the much better time you will have deploying. Skip to the next section if you will not be using frame packs and can park a vehicle within a short walking distance to the deployment site. This section is only for deployments that must be hiked through vegetation to be deployed.

Pack your frame packs with heavier items closer to your back and attempt to achieve an even distribution of weight across both people, depending on the capabilities of each person. Have each person walk around with their packs beforehand to ensure it's a reasonable weight. Our builds should not require a total weight heavier than 100lbs, including the frame packs.



4.2 Set-up

- 1) Lay out the solar panels, and place the battery and the box beneath them for sun coverage.
- 2) Loop 3 guylines around the top before extending higher, stopping a few inches before the maximum height.

- 3) Attach the battery cables to the battery, and ensure the device turns on. This could be a good moment to set the solar charge controller to “Lithium” if you haven’t already done so.
- 4) Attach the solar cables next, and ensure the solar manager sees them.
- 5) Fill your weight bags with sand/dirt nearby and place them on the bottom legs of the tripod, securing them together with the carabiner or zip ties.
- 6) With each guy line, pull it out a short distance from the tripod, ensuring all 3 form a semi-equidistant triangle to support the tripod from all 3 sides.
- 7) Drill pilot holes if necessary, and drill in the ground stake with the turnbuckles in those 3 spots, pulling the guyline to ensure that it’s tight but there is some slack still.
- 8) Loop the guy line to each turnbuckle, and tighten all 3 as needed to remove slack from the guylines.
- 9) Once all 3 are tight and it feels secure, raise the tripod additionally to remove any left-over slack. The pole should feel very secure with each guyline taught.
- 10) Do the same for the other tripod if using separate tripods for the camera and Nanobeam.
- 11) Secure the AudioMoth to the base of the nearest tripod, only a few feet off the ground.



4.3 Final Checks

- 1) Connect to the Nanobeam Wifi and go to the UISP app. You should be able to log-in to the Nanobeam.
- 2) See if the device connected to the remote nanobeam, and use the aiming tool to move the Nanobeam for a better connection.
- 3) Zip tie any loose cables, and ensure the box is locked, and the battery is sealed in the bag.
- 4) Nip the excess zip tie off and gather trash. Tighten any turnbuckles.